
Session 3 — Analysis I

Exercise sheet

Functions, derivatives, integrals, and a first differential equation. Two examples recur throughout: the forgetting curve $R(t) = R_0 e^{-t/\tau}$, giving the proportion of a word list still recalled after a delay t , and the learning curve $\dot{P} = k(P_{\max} - P)$, describing performance approaching a ceiling with practice. Show your reasoning — most answers are two or three lines — and check every integral by differentiating it and every solution by substituting it.

Exercise 1 Growth and decay.

- (a) For each, say whether it grows or decays, and give the time constant τ where it decays: e^{3t} , e^{-3t} , $e^{-t/5}$, $4e^{-t/0.5}$.
 - (b) Write $e^{-t/5}$ in the form e^{at} and give a .
 - (c) Simplify $\ln(e^5)$, $e^{\ln 7}$, $\ln(a^2b)$, $\ln(e^{-t/\tau})$.
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Exercise 2 Recall of a learned word list decays as $R(t) = R_0 e^{-t/\tau}$ — the shape Ebbinghaus measured in 1885.

Take $R_0 = 0.8$ and $\tau = 3$ days.

- (a) Compute $R(0)$, $R(3)$ and $R(9)$ using $e^{-1} \approx 0.37$, $e^{-3} \approx 0.05$.
 - (b) After how long has recall halved? (Solve $e^{-t/\tau} = \frac{1}{2}$ and give the answer as a multiple of τ .)
 - (c) Two participants have $\tau = 2$ and $\tau = 8$ days. After 8 days, who remembers more, and roughly how much does each have left?
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Exercise 3 The power law of practice states $RT(n) = a n^{-b}$, with n the number of practice trials.

- (a) With $a = 800$ ms and $b = 0.3$, compute $RT(1)$ and $RT(100)$.
 - (b) Show that $\ln RT = \ln a - b \ln n$. What does this say about the shape of the data on log-log axes?
 - (c) In one sentence: how would you tell a power law from an exponential, given a data set?
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Exercise 4 Differentiate.

- (a) $f(t) = t^2 - 4t + 7$; $g(t) = 5e^{-2t}$; $h(t) = 1/t + \sqrt{t}$; $k(t) = 3 \ln t$.

- (b) $R(t) = R_0 e^{-t/\tau}$. Show that $R' = -R/\tau$, and say in words what that equation asserts.
- (c) $RT(n) = 800 n^{-0.3}$. Is RT' positive or negative, and what happens to its size as n grows?
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Exercise 5 Chain rule and product rule.

- (a) $f(t) = e^{-t^2}$; $g(t) = \ln(1 + 3t)$; $h(t) = e^{-(t-2)/5}$.
- (b) $f(t) = t e^{-t}$. Find where $f' = 0$ and decide whether it is a maximum or a minimum. Sketch the curve.
- (c) $f(t) = (1 + t^2)^3$.
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Exercise 6 Someone writes $\frac{d}{dt}e^{-t/2} = \left(-\frac{t}{2}\right)e^{(-t/2)-1}$.

- (a) Which rule have they applied, and why does it not apply here?
- (b) Give the correct derivative.
- (c) State the one question that prevents this error.
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Exercise 7 Using signs only.

- (a) For $f(t) = t^3 - 3t$, find where $f' = 0$ and classify each point from the sign of f' on either side.
- (b) Show that $f(t) = t^3$ has $f'(0) = 0$ but no maximum and no minimum there.
- (c) $P(t) = 100 - 80 e^{-t/10}$. Compute P' and show it is positive for all t . What is $\lim_{t \rightarrow \infty} P(t)$?
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Exercise 8 Compute directly, using the table of antiderivatives.

- (a) $\int_0^2 (3t^2 - 4t + 1) dt$; $\int_1^e \frac{dt}{t}$; $\int_0^1 \sqrt{t} dt$.
- (b) $\int_0^\infty e^{-3t} dt$. Interpret the answer.
- (c) $\int_0^T e^{-t/\tau} dt$, and its limit as $T \rightarrow \infty$.
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Exercise 9 By substitution.

- (a) $\int \frac{2t}{1+t^2} dt$;
- (b) $\int 2t e^{t^2} dt$;

(c) $\int_0^1 t e^{-t^2} dt.$

Check each answer by differentiating it.

Exercise 10 By parts.

(a) $\int t e^{-2t} dt;$

(b) $\int_0^\infty t e^{-t} dt;$

(c) $\int \ln t dt.$ (Hint: take $u = \ln t$, $v' = 1$.)

Exercise 11 A probability density on $[0, \infty)$ is a function $p \geq 0$ with $\int_0^\infty p(t) dt = 1$; the mean of the distribution is $\int_0^\infty t p(t) dt$. Every expected value in the statistics sessions is an integral of this kind.

Take $p(t) = \lambda e^{-\lambda t}$ with $\lambda > 0$.

(a) Verify that $\int_0^\infty p(t) dt = 1$.

(b) Compute the mean, and express it in terms of the time constant.

(c) Integrate the forgetting curve $R_0 e^{-t/\tau}$ over all $t \geq 0$ and interpret the result.

Exercise 12 For each, say whether it is an integration problem or a differential equation proper, then solve it with $x_0 = 1$.

(a) $\dot{x} = 3t^2;$

(b) $\dot{x} = -2x;$

(c) $\dot{x} = -2x + 5;$

(d) $\dot{x} = e^{-t}.$

Exercise 13 Someone proposes $x(t) = -x^2/2 + 3x$ as the solution of $\dot{x} = -(x - 3)$.

(a) Give the one-line “type” argument showing it cannot be a solution.

(b) Give the substitution argument: differentiate it with respect to t using the chain rule and show the equation is not satisfied.

(c) Give the correct solution with $x(0) = x_0$.

Exercise 14 The equation $\dot{x} = ax + b$.

- (a) Verify by substitution that $x(t) = x^* + (x_0 - x^*)e^{at}$ solves it, with $x^* = -b/a$.
 - (b) For $\dot{x} = -x + 4$: give x^* and τ , decide stability, and describe the trajectory from $x_0 = 10$ and from $x_0 = 0$.
 - (c) For $\dot{x} = 3x - 6$: give x^* , decide stability, and describe what happens from $x_0 = 1.9$ and from $x_0 = 2.1$.
 - (d) What goes wrong if $a = 0$? Solve $\dot{x} = b$ directly.
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Exercise 15 *Performance improves toward a ceiling, and the closer you get the slower you improve: $\dot{P} = k(P_{\max} - P)$. This is $\dot{x} = ax + b$ in disguise.*

- (a) Identify a and b , and give P^* and τ .
 - (b) With $P_0 = 20$, $P_{\max} = 100$ and $k = 0.1$ per session, write $P(t)$ and compute $P(10)$ and $P(30)$.
 - (c) How many sessions are needed to close 95% of the gap?
 - (d) Show that the forgetting curve satisfies the same equation with $b = 0$, and say in one sentence what the two curves have in common.
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Exercise 16 Phase line and stability.

- (a) Draw the phase line for $\dot{x} = -x + 3$ and for $\dot{x} = x - 3$: mark x^* and put an arrow on each side.
 - (b) Explain, without solving, why the first is stable and the second is not.
 - (c) A quantity obeys $\dot{x} = ax + b$ and is observed to settle at $x = 7$ with a time constant of 4. Give a and b .
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Exercise 17 Euler's method.

- (a) Take two steps of $\dot{P} = 0.1(100 - P)$ from $P_0 = 20$ with $\Delta t = 1$. Compare with the exact value $P(2)$.
- (b) Take one step of $\dot{x} = -x + 2$ from $x = 0$ with $\Delta t = 0.1$.
- (c) For $\dot{x} = -x/5$, what is the largest step that does not blow up?
- (d) In one sentence: why is “the simulation did not explode” not the same as “the simulation is accurate”?